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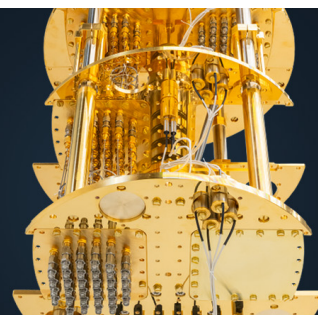
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Aromatic poly(arylene ether urea) with high dipole moment for high thermal stability and high energy density capacitors

Zhaoxi Cheng,^{1,2} Minren Lin,¹ Shan Wu,¹ Yash Thakur,¹ Yue Zhou,¹ Dae-Yong Jeong,³ Qundong Shen,² and Q. M. Zhang^{1,a)}

¹Department of Electrical Engineering and Materials Research Institute, The Pennsylvania State University, University Park, Pennsylvania 16802, USA

²Department of Polymer Science and Engineering and Key Laboratory of High Performance Polymer Materials and Technology of MOE, School of Chemistry and Chemical Engineering, Nanjing University, Nanjing 210093, China

³Department of Materials Science and Engineering, Inha University, Incheon 402-751, South Korea

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Developing dielectric polymers with higher dielectric constant without sacrificing loss and thermal stability is of great importance for next generation of high energy density capacitors. We show here that by replacing the CH₂ group in the aromatic polyurea (ArPU) with the polar ether group, thus raising the dipole moment of the molecular unit, poly(arylene ether urea) (PEEU) shows an increased dielectric constant of 4.7, compared with 4.2 of ArPU. Moreover, PEEU maintains the low dielectric loss and is thermally stable up to 250 °C. As a result, the polymer delivers 13 J/cm³ discharged energy density at room temperature and 9 J/cm³ at 120 °C. The high quality films perform well in terms of both breakdown strength (at 700 MV/m at room temperature) and leakage current from room temperature to elevated temperature. At 120 °C, the breakdown strength is 600 MV/m and the conductivity is 1.58×10^{-14} S/cm measured under 100 MV/m. © 2015 AIP Publishing LLC. [<http://dx.doi.org/10.1063/1.4921485>]

Dielectric capacitors with high energy density, low loss, and high operating temperature are desired for a broad range of applications such as hybrid electrical vehicles (HEVs), pulse power weapon systems, and switch mode power supplies.^{1–5} Compared with ceramic capacitors, polymer-film capacitors possess several advantages such as high dielectric strength, high energy density and low dielectric loss. However, for the traditional polymer capacitors such as biaxially oriented polypropylene (BOPP), the dielectric constant is low ($K \approx 2.2$), the energy density is limited to 5 J/cm³, and the maximum operating temperature is lower than 100 °C.³ Low energy density leads to significant volume and hinders the miniaturization of electronic devices. In addition, low operating temperature limits the broad applications of capacitors. In many of these widely used linear dielectrics like BOPP, conduction loss becomes more significant at higher applied fields.^{6,7} Usually, these losses increase exponentially with the electric field, and cause Ohmic heating of the capacitors.^{2,6–8} This results in the need to have a cooling system to avoid overheating the BOPP film capacitors. For example, in hybrid electric vehicles, an extra cooling loop has to be introduced in the BOPP capacitor banks in order to prevent a runaway temperature increase caused by the conduction loss heating.

In an effort to increase the energy density, PVDF-based polymers with high dielectric constant were developed which showed 25 J/cm³ energy density.^{4,9,10} However, these polar polymers with strongly coupled dipoles exhibit pronounced polarization hysteresis at high electric fields which leads to high loss.

Several studies have been conducted earlier on ArPU for dielectric applications due to its relatively high dielectric constant (~ 4.2), low loss ($\sim 1\%$), and high thermal stability (>150 °C).^{11–15} In this letter, we show that replacing the CH₂ group in aromatic polyurea (ArPU) (Fig. 1), by the polar ether group, can lead to an increase in the dielectric constant, to 4.7, without sacrificing the loss and thermal stability. Thus, the results presented here suggest an effective approach of developing dielectric polymers with high dielectric constant, low loss, and high thermal stability by designing and developing highly dipolar polymers with high dipole moments, accompanied by benzene rings for thermal stability and low loss.

Poly(arylene ether urea) (PEEU) was synthesized from (*m*-phenylenedioxy) dianiline and diphenyle carbonate by thermal polycondensation as shown in Fig. 1(b). The mixture of the two monomers was stirred at 150 °C in a vacuum for 4 hours, and PEEU powder was obtained through purification by washing with ethanol for 5–6 times. There is no solvent and no catalyst used in the synthesis. It is a green and low cost thermal polycondensation process. The average molecular weight (*M_n*) is in the range of 2000–2500. PEEU films were prepared by dissolving the powders into dimethyl formamide (DMF) at 65 °C, and solution was casted onto glass plates and held at 80 °C for 22 hours to remove the solvent. The films obtained were annealed at 150 °C for 2 days to further remove the solvent. Gold (Au) film was sputtered as electrodes for the electrical characterization.

The x-ray diffraction of PEEU films was carried out at room temperature using Panalytical Xpert Pro MPD diffractometer. The thermal properties were characterized with thermal gravimetric analysis (TGA, 2050, TA), and differential scanning calorimeter (DSC, Q100, TA) at the heating/cooling

^{a)} Author to whom correspondence should be addressed. Electronic mail: qxz1@psu.edu.

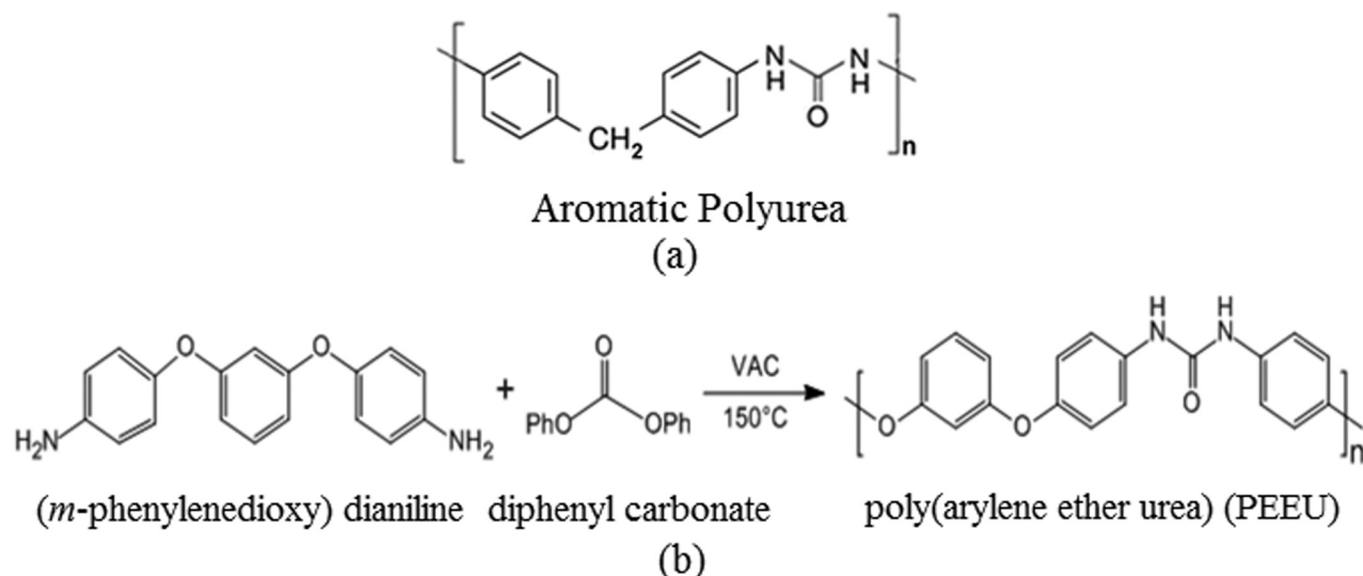


FIG. 1. (a) Schematic of chemical structure of aromatic polyurea (ArPU). (b) Schematic of synthesis and chemical structures of poly(arylene ether urea).

rate of 10 °C/min in nitrogen atmosphere. The dielectric properties and AC conductivity measurement were conducted with HP 4294 precision impedance analyzer. The HP 4284 LCR meter and Delta 9023 environment chamber were assembled to study the dielectric properties vs temperature. The polarization–electric field loops (P-E loops) and breakdown strength were determined in the range from room temperature to elevated temperature with a Sawyer-Tower circuit.

The XRD data of the PEEU films, presented in Fig. 2(a), do not show any sharp peaks, which suggests that the PEEU film has an amorphous structure. The TGA and DSC data of the PEEU films are presented in Figs. 2(b) and 2(c). Fig. 2(b) reveals that the films are thermally stable up to 220 °C, and just show a small weight loss of 0.4% at 220 °C. The rate of losing weight reached a maximum, 1.1%/°C at 353 °C. The thermal properties, including initial decomposition temperature, the temperature range of significantly losing weight, and the remaining weight, were similar with other derivatives of aromatic urea-based polymers.^{14,15} The DSC data in Fig. 2(c) do not show any melting peak, which is consistent with the amorphous nature of PEEU, and glass transition in the temperature range measured. The broad peak of heat absorption above 300 °C is due to the decomposition of PEEU. Here, it should be noted that the benzene rings

introduced to the main chain improve the thermal stability of PEEU.¹⁶

Fig. 3 presents the dielectric properties of PEEU as a function of frequency and temperature. Due to high dipole moment,¹⁷ PEEU displays a room temperature dielectric constant of 4.7 and loss of 1.1% measured at 1 kHz, as seen in Fig. 3(a). Both the dielectric constant and loss do not change with frequency. The dielectric constant 4.7 of PEEU is larger than 4.2 of ArPU.¹⁴ The large dielectric constant of PEEU is due to the polar ether group replacing the CH₂ group in ArPU. Similar to the frequency response, the dielectric properties do not change with temperature, measured at 1 kHz from room temperature to 200 °C (Fig. 3(b)), showing thermal stability.

The energy storage properties at room temperature and elevated temperature are presented in Fig. 4. At room temperature, PEEU films show the behavior of a linear dielectric with low loss for dielectric breakdown of 700 MV/m (Fig. 4(a)). The discharge energy density is 13 J/cm³ as showed in Fig. 4(b), which is higher than that of aromatic polyurea.¹⁴ The charge-discharge efficiency of over 95% is achieved for fields below 500 MV/m. At high temperatures, the PEEU films performed well in the whole temperature range (up to 120 °C). The breakdown field as a function of temperature, based on the two-parameter Weibull analysis, is presented in

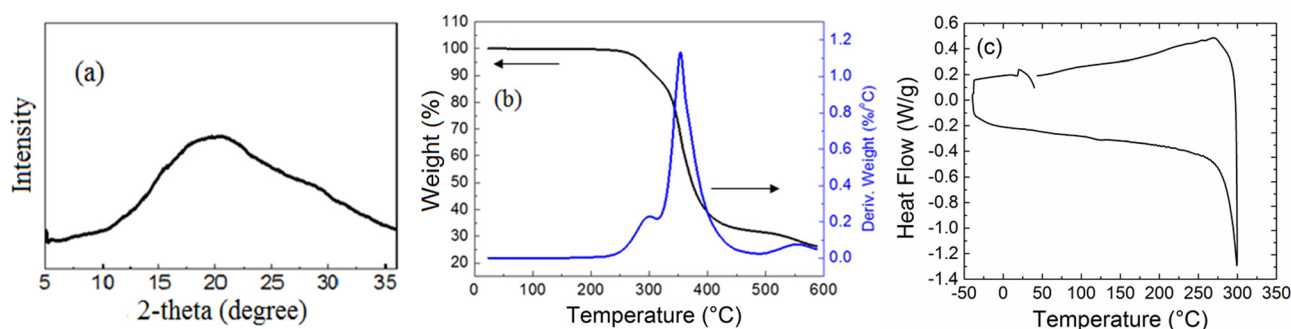


FIG. 2. (a) X-ray data for PEEU film, (b) TGA, and (c) DSC vs. temperature. The heating rate is 10 °C/min.

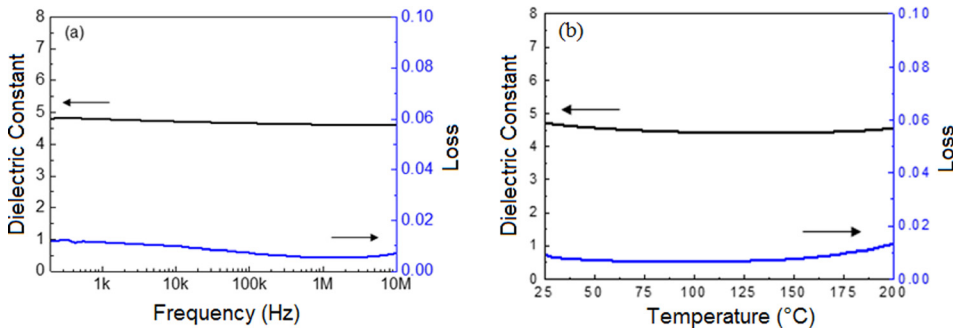


FIG. 3. The dielectric constant and dielectric loss as a function of (a) frequency at room temperature and (b) temperature at 1 kHz.

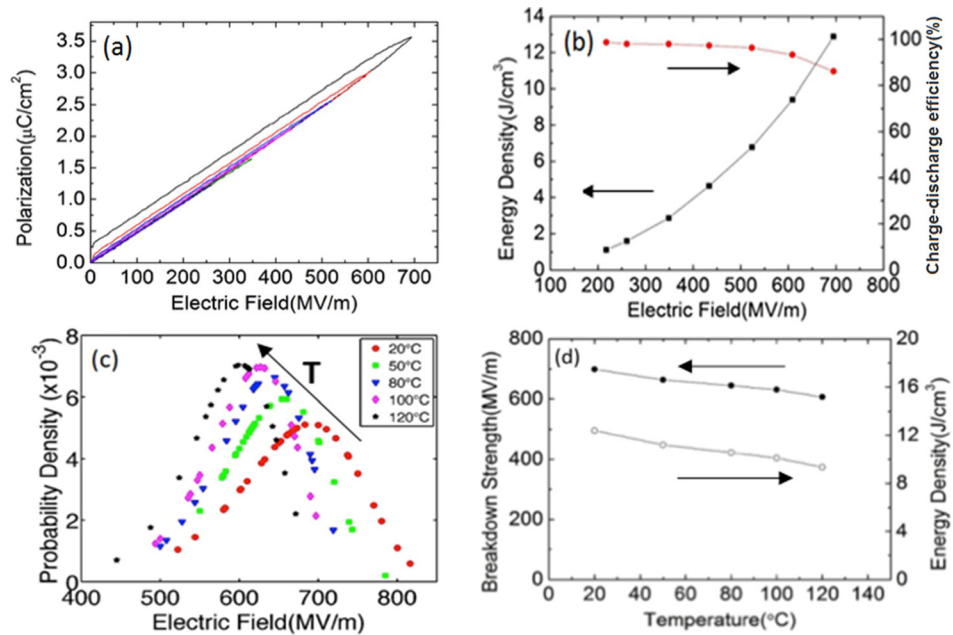


FIG. 4. (a) Polarization-electric field data (P-E Loops) of PEEU films at 10 Hz, room temperature, (b) energy density/charge-discharge efficiency versus electric field, (c) Weibull distribution of breakdown field, (d) temperature dependence of breakdown field/energy density of PEEU film in the temperature range from 20 °C to 120 °C.

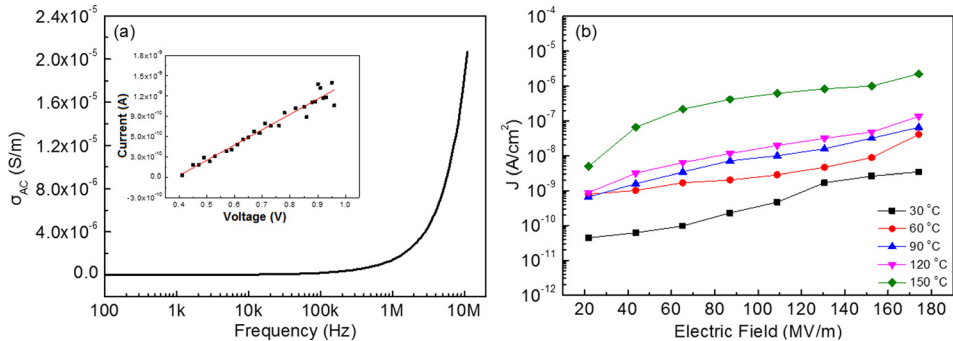


FIG. 5. (a) Frequency dependence of AC conductivity at room temperature, inset shows the I-V curve for determining the DC conductivity, (b) current density-electric field at different temperatures.

Fig. 4(c). Both the breakdown field and energy density decrease slightly with temperature, as shown in Fig. 4(d).

Since conduction loss is a major issue for dielectric polymers at high electric field and temperature,^{18–22} the alternating current (AC) and direct current (DC) conduction measurements were carried out with applied electric field and temperature to investigate the conduction of the polymer films. The total conductivity can be expressed as $\sigma(\omega, T) = \sigma_{AC}(\omega, T) + \sigma_{DC}(T)$, where σ_{AC} is the AC conductivity, and σ_{DC} is the DC conductivity. The AC conductivity of PEEU film at room temperature is presented in Fig. 5(a), which could be fitted by a power law, $\sigma \propto \omega^n$, $n=1$. It obeys the universal dynamic response (UDR) which is due to the hopping effect of localized charge and $\sigma_{AC} = \epsilon_0 \epsilon_r \omega \tan \delta$, where ϵ_r is the relative dielectric constant, ϵ_0 is the vacuum permittivity, and $\tan \delta$ is

the low tangent.^{18,19} For hopping conduction, the DC conductivity, which is temperature dependent, is determined by the activation energy of carrier hopping through the sample.²⁰ σ_{DC} was obtained from I-V curve as shown in the inset of Fig. 5(a), which follows the Ohm's law at low applied field and is equal to σ_{AC} at frequencies below 100 kHz. Fig. 5(b) presents the current density J-electric field E curves vs applied field over a broad temperature range, and the observed behavior is similar to that of the irradiated polyimide films.^{21,22} The current density at 30 °C is 3.6×10^{-10} A/cm² at 100 MV/m. The small conduction current at temperature below 120 °C results in a low conduction loss and results in high efficiency energy storage at high temperature.

In conclusion, an amorphous aromatic polymer, PEEU, was developed which is thermally stable up to 250 °C. The

experimental results show that by replacing CH₂ in ArPU by the more polar ether group, the dielectric constant is increased to 4.7, compared with 4.2 of ArPU. The higher dielectric constant leads to a high discharge energy density of 13 J/cm³ at 700 MV/m, and the charge/discharge efficiency of 95% at 500 MV/m. The high charge/discharge efficiency is a result of low conduction loss of the PEEU films in the temperature range from room temperature to 120 °C. The conductivity at 100 MV/m is 3.45×10^{-16} S/cm and 1.58×10^{-14} S/cm, at 30 °C and 120 °C, respectively, better than the conductivity of BOPP at room temperature, which is 3.8×10^{-14} S/cm at 100 MV/m.

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